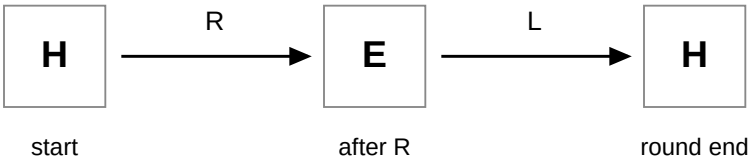
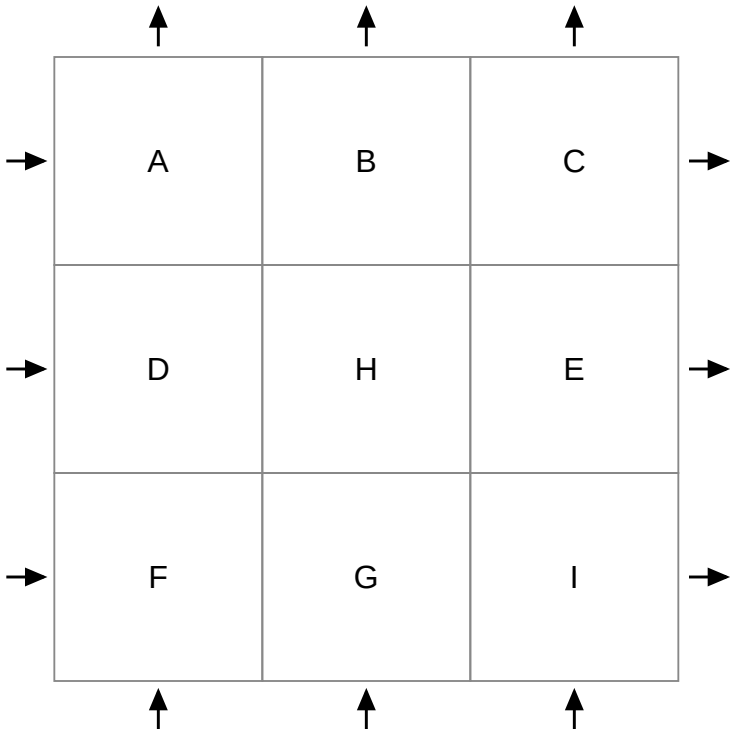
R, L, U, D mean one cell right, left, up, down. On the 3-by-3 board, cross an edge to the opposite edge in the same row or column. One word is a whole round. Repeated maps continue beyond the page; their borders do not wrap.

RL



Problem 1: Start at H. Repeat each word without resetting your pawn. Mark the cell after each whole round. Which words reach every cell at round ends? Make words with different repeating patterns.



R

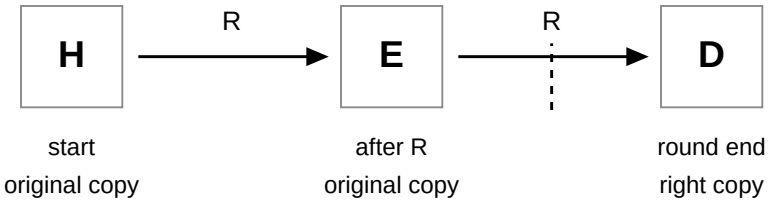
RU

RRR

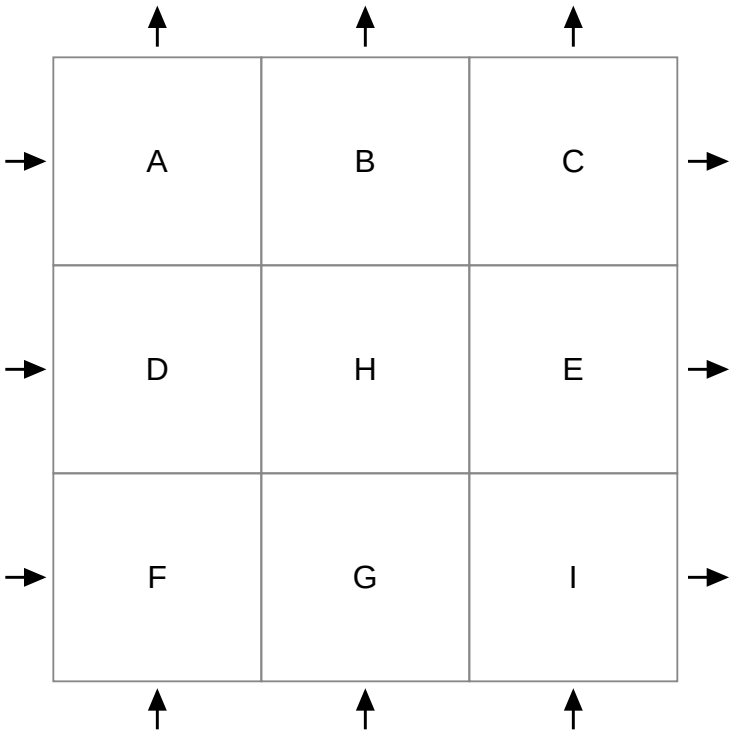
RULD

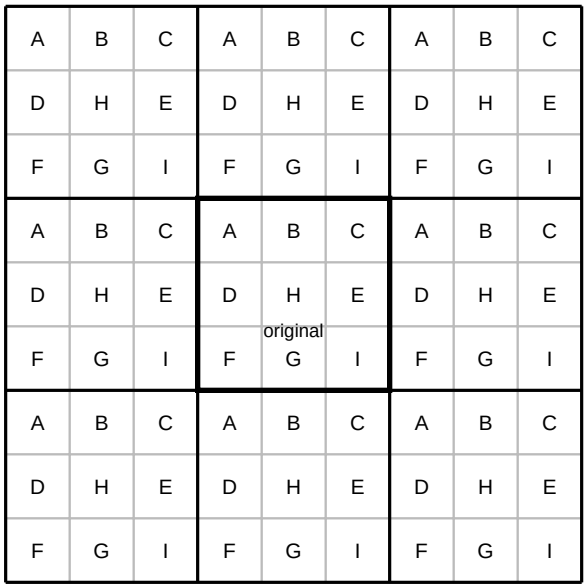
RRU

RR



Problem 2: Make a nine-step trip from H that visits every cell once before returning to H. Find tours with different finishing copies of H on the repeated map. Could a nine-step tour finish in the original copy?





Problem 3: B and E are closed in every copy. From the original H, reach each circled H in as few steps as possible. Keep a move word for each route. What makes a shorter route impossible?

C	A	X	C	A	X	C
X	D	**H** up copy	X	D	**H** right + up	X
I	F	G	I	F	G	I
C	A	X	C	A	X	C
X	D	**H** original	X	D	**H** right copy	X
I	F	G	I	F	G	I
C	A	X	C	A	X	C

right copy _____

up copy _____

right + up _____